

2 The patty pan squash plant, *Cucurbita pepo*, produces edible fruits that vary in colour.

- (a) The colour of the fruits is controlled by two genes, **A/a** and **B/b**, that occur on different chromosomes.
- Allele **A** produces a white fruit colour.
 - Allele **a** does not produce a colour by itself but allows the colours coded by gene **B/b** to show in the phenotype.
 - Allele **B** produces a yellow fruit colour.
 - Allele **b** produces a green fruit colour.

In a dihybrid cross, an **AABB** plant was crossed with an **aabb** plant. All the resulting F1 plants produced white fruits.

The F1 plants were then crossed with each other to obtain the F2 generation.

- (i) Complete Fig. 2.1 with the gametes produced by the F1 parents, the F2 genotypes and the F2 phenotypes.
- State the ratio of fruit colours in the F2 offspring.

F1 parents:	AaBb	×	AaBb	

Fig. 2.1

ratio of fruit colours in F2 offspring [4]

- (ii) Test crosses were carried out on two white-fruited plants, **P** and **Q**, from the F₂ generation. Each of these plants had its female flowers pollinated with pollen from a green-fruited plant.

For plant **P**, half of the offspring were white and half were yellow.
For plant **Q**, half of the offspring were white and half were green.

Deduce the genotypes of plants **P** and **Q**.

plant **P**

plant **Q**

[2]

- (iii) Plants **P** and **Q** show genetic variation with respect to fruit colour alleles.

Identify the process that occurred during meiosis in the F₁ parents that produced this variation **and** the stage of meiosis at which it occurred.

process

stage of meiosis

[2]

- (b) In 1994, two new varieties of squash were grown in the USA. Both varieties showed resistance to two viral diseases of squash plants.

- The variety 'Tigress' was developed by selective breeding.
- The variety 'Freedom II' was developed by adding genes for viral coat proteins to the squash genome.

- (i) Identify, with reasons, the variety of squash that can be described as recombinant.

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..... [2]

- (ii) Outline **one** social advantage and **one** social disadvantage of the recombinant squash variety compared to non-resistant squash varieties.

advantage
.....
disadvantage
..... [2]

[Total: 12]